

A

A



1

A

10 9 8 7 6 5 4 3

INTERFACE KEY DETAIL

Project: VARIOUS

2064000-TS
(ALL KEYS SUPERIMPOSED)

4064000-TS
(ALL KEYS SUPERIMPOSED)

6064000-TS
(ALL KEYS SUPERIMPOSED)

TYP. 2.4 ±0.1 (LP ONLY)

TYP. 0.2 ±0.1 (LP ONLY)

2x 4.1 ±0.1

TYP. 1.0 ±0.1 (LP ONLY)

2x 6.1 ±0.1

4x 2
4x 1

KEY A

KEY B

KEY A

KEY B

KEY A

KEY B

KEY C

KEY D

KEY C

KEY D

KEY C

KEY D

SECTION PLANE 4
SCALE: 2/1

SECTION PLANE 4
SCALE: 2/1

SECTION PLANE 4
SCALE: 2/1

TYP. 3.5 (064 ONLY)

2280000-TS
(ALL KEYS SUPERIMPOSED)

4280000-TS
(ALL KEYS SUPERIMPOSED)

6280000-TS
(ALL KEYS SUPERIMPOSED)

TYP. 4.55 ±0.10 (HP ONLY)

8x 1.4 ±0.1
8x 0.8 ±0.1
8x 4.2 ±0.1

4x 6.5 ±0.1
0.3 ±0.1

TYP. 1.2 ±0.0 (HP ONLY)

4x 2.0 ±0.1

8x 45° ±1°

KEY A

KEY B

KEY A

KEY B

KEY A

KEY B

KEY C

KEY D

KEY C

KEY D

KEY C

KEY D

SECTION PLANE 4
SCALE: 2/1

SECTION PLANE 4
SCALE: 2/1

SECTION PLANE 4
SCALE: 2/1

TYP. 1.2 ±0.1 (HP ONLY)

TYP. 2.45 (280 ONLY)

NOTE:
ALL CENTERLINE PLANES ESTABLISHED BY DATUM B
ALL RIBS 0.5 DEGREE DRAFT ANGLE (PLANE 4 NEUTRAL PLANE)
ALL DIMENSIONS ON PAGE: ± 0.2 A B

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ITEM NAME		LOW VOLTAGE CONNECTOR STANDARD	
ITEM NUMBER	DRW-00456412	REVISION	SHEET
		C.5	2 of 12

TYP. 15.8 (HP ONLY) MIN.
TYP. 12.8 (HP ONLY) MIN.

INTERFACE	PLANE 1	PLANE 2	PLANE 4
2280000-TS	16.11	*16.50	16.62
4280000-TS			
6280000-TS			

INTERFACE	PLANE 1	PLANE 2	PLANE 4
2280000 - TS	16.01	*16.40	16.52
4280000 - TS	25.61	*26.00	26.12
6280000 - TS			

INTERFACE	PLANE 1	PLANE 2	PLANE 3	PLANE 4	PLANE 5
2280000 - TS	14.76	*13.90	13.81	11.98	*11.90
4280000 - TS					
6280000 - TS					

INTERFACE	PLANE 1	PLANE 2	PLANE 3	PLANE 4	PLANE 5
2280000-TS	14.66	*13.80	13.71	11.88	*11.80
4280000-TS	24.26	*23.40	23.31	21.48	*21.40
6280000-TS					

3.0 +Y (PLANE 2) MIN.
3.0 +Y (PLANE 2) MIN.

ACTUAL OR THEORETICAL
MEASUREMENT POINT

4.0

—

TYP. 3.0 MAX.

1.6 MAX.
NOT TO EXCEED ABOVE
MEASURING POINT
UNCONTROLLED BELOW
MEASURING POINT

ACTUAL OR THEORETICAL
MEASUREMENT POINT

4.0

AS MEASURED FROM
TOP OF INTERFACE 5.5 MIN.

HEADER KEEP OUT ZONE AND POLARIZATION FEATURE
SCALE: 3/1

3.7 ± 0.1 —

	0.2 (M)	A	B
---	---------	---	---

PLANE 2 ▼

▼ PLANE 2

PLANE 5 ▼

▼ PLANE 5

FRONT VIEW
SCALE: 3/1

REFER TO EWCAP 064-T (HYBRID ONLY)

	0.3 (M)	A	B
---	---------	---	---

REFER TO EWCAP 064-T (HYBRID ONLY)

— TYP. 5.4 ± 0.1
 — TYP. $1.2^{+0.4}_{-0.2}$

	0.3		A	B
--	-----	--	---	---

TYP. REFER TO EWCAP 280-T001

ALL CORNERS CONCENTRIC
R 2.7

 9.95 ± 0.20

TYP. REFER TO EWCAP 280-T00

	0.3	(M)	A	B
--	-----	-----	---	---

SECTION PLANE 2

SCALE: 4/1

TRUE POSITION MEASURED AT TOP OF TERMINAL

SECTION PLANE 5

SCALE: 4/1

TYP. REFER TO EWCAP 280-T001

	0.2	(M)	A	B
---	-----	-----	---	---

MEASURED AT DATUM A

TOP VIEW
SCALE: 4/1

RIBS IGNORED IN PROFILE MEASUREMENT

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ITEM NAME

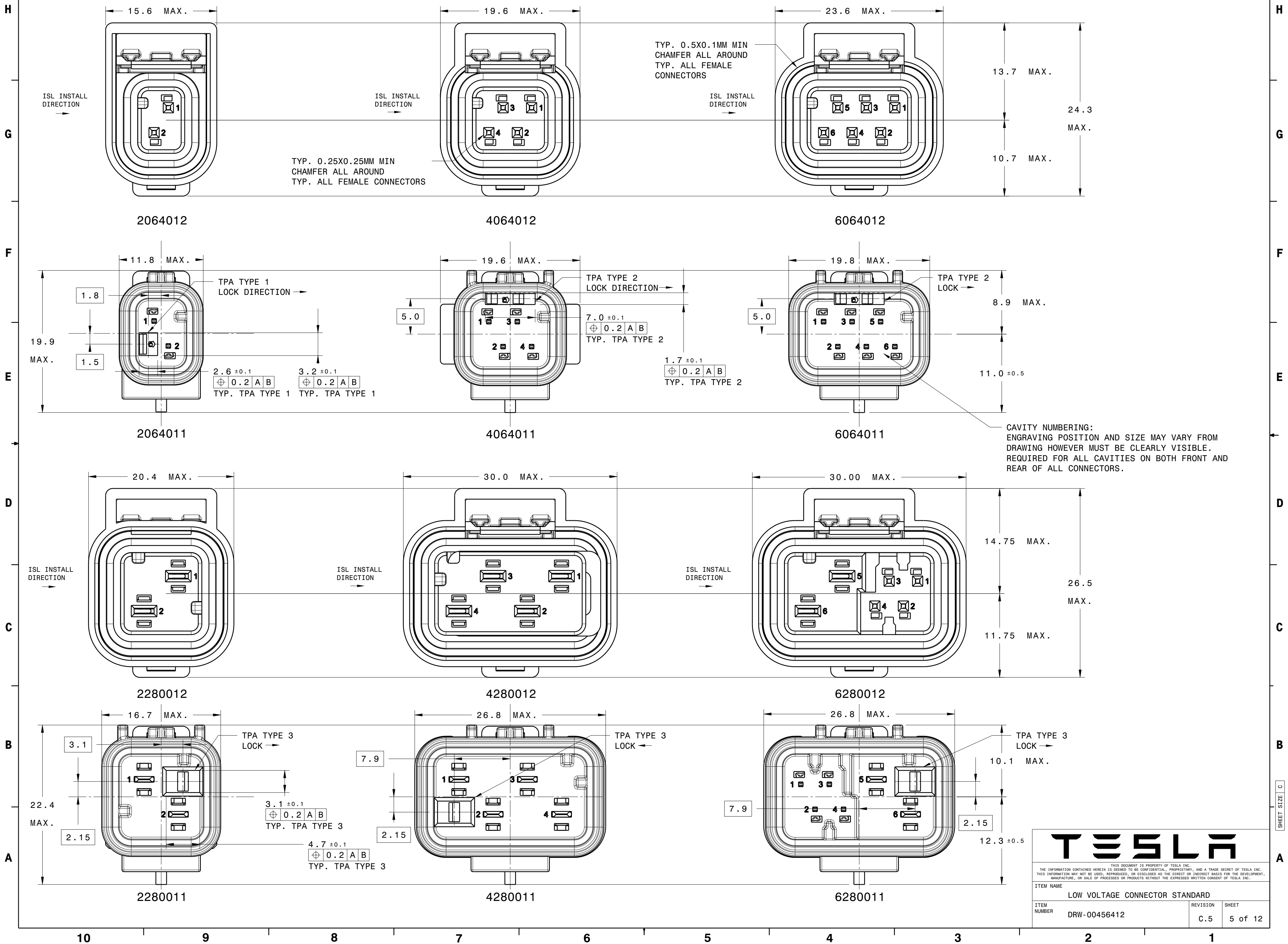
LOW VOLTAGE CONNECTOR STANDARD

ITEM NUMBER	DRW-00456412
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REVISION	SHEET
0.5	1 of 10

FRONT CONNECTOR VIEW

MAXIMUM EXTERNAL DIMENSIONS SHOWN TO ENSURE PACKAGING. ANY TEST OR ASSEMBLY FIXTURES MUST LOCATE TO CONTROLLED INTERFACE, NOT EXTERNAL DIMENSIONS.

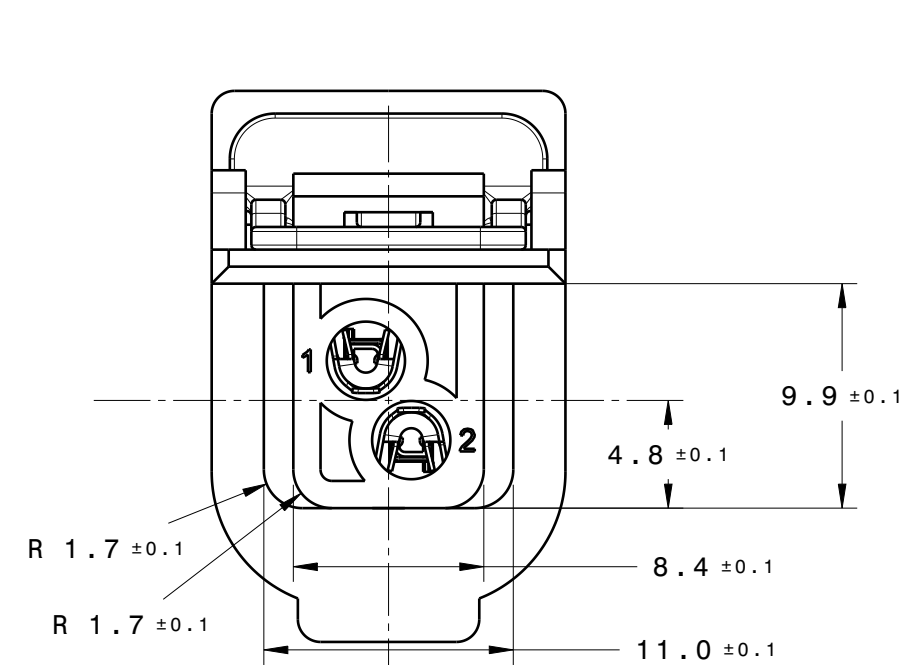


10 9 8 7 6 5 4 3

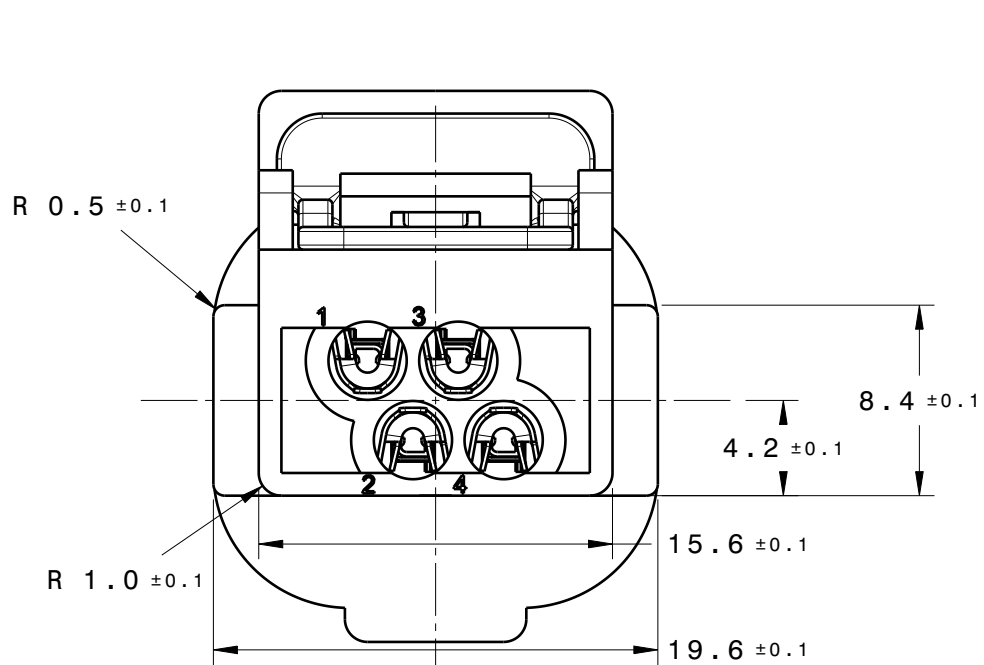
REAR CONNECTOR VIEW

CABLE EXIT AND ACCESSORY RIB FEATURE PRECISION TO ENSURE COMPATIBILITY

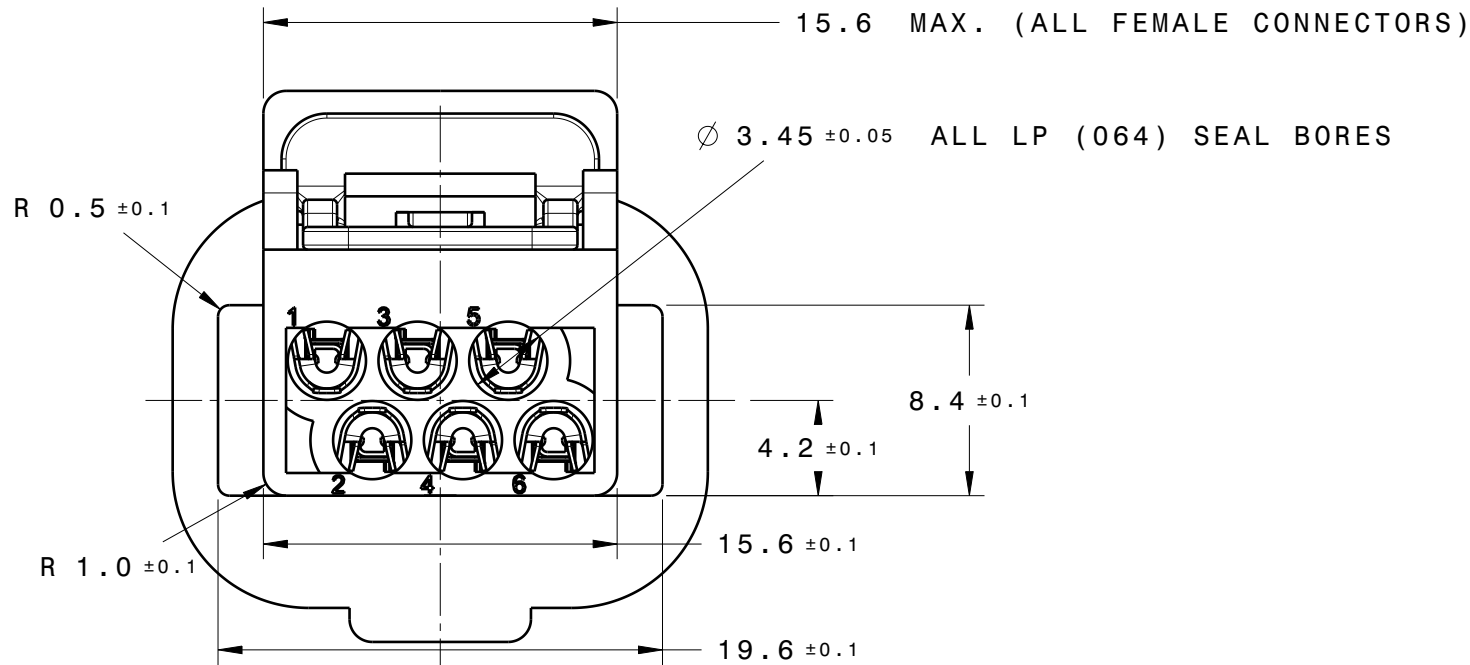
Project: VARIOUS



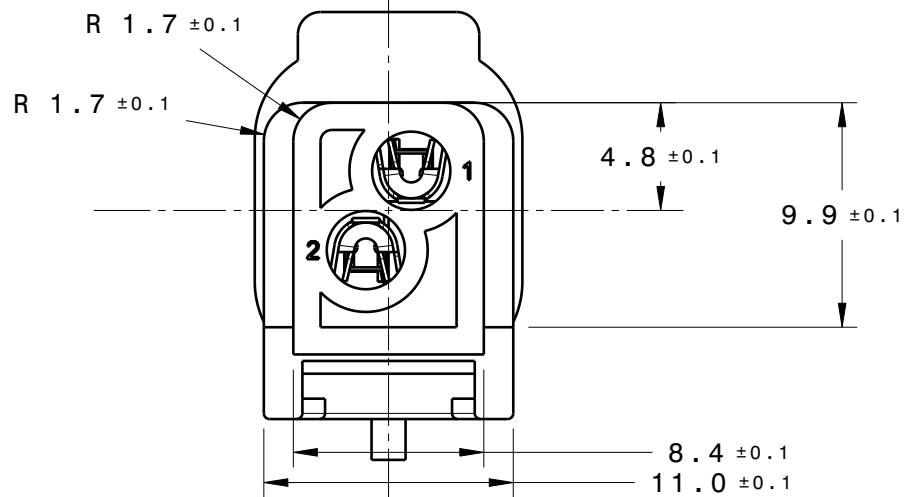
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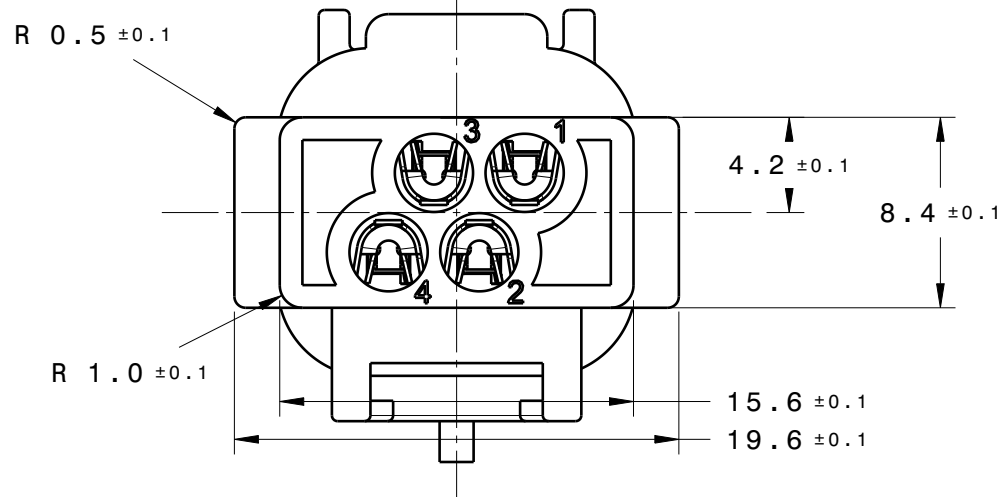
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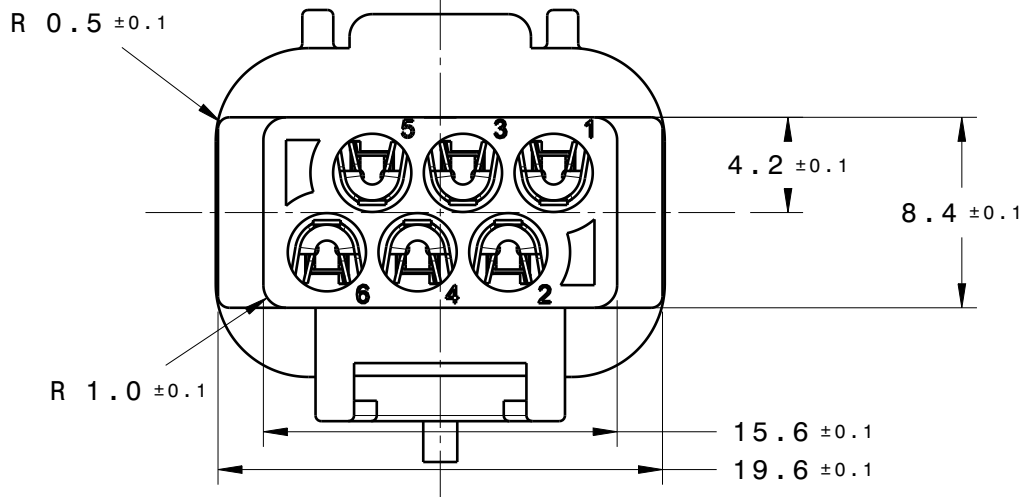
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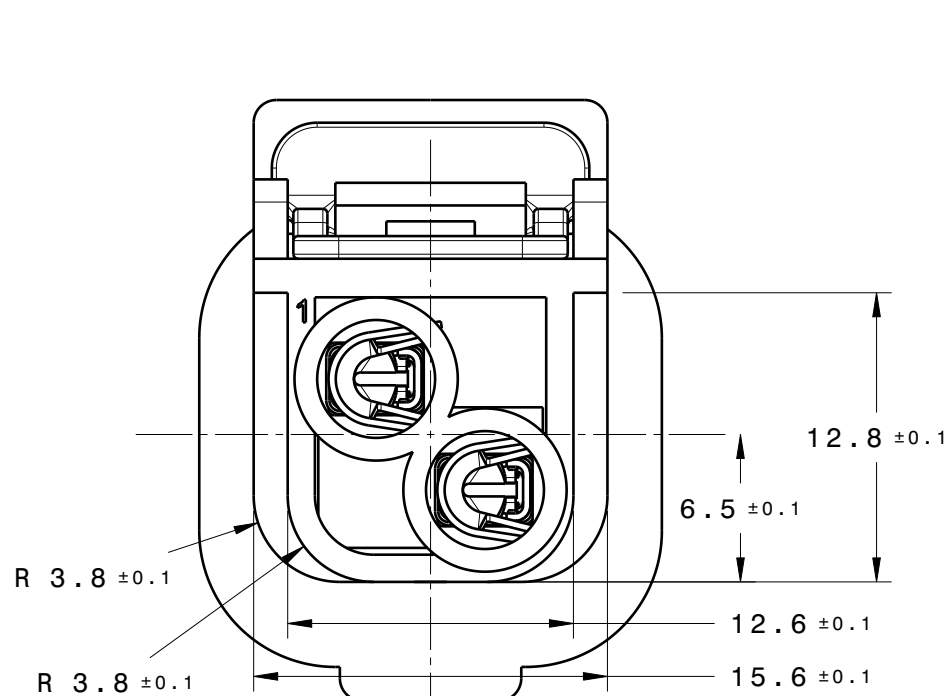
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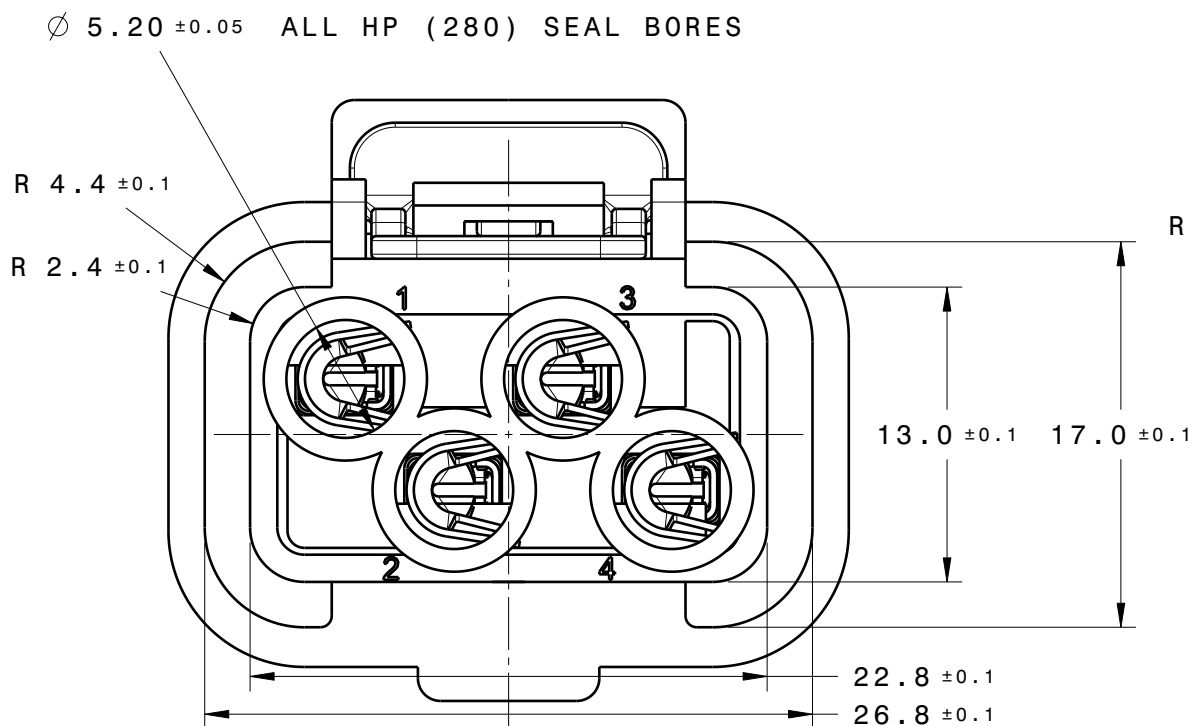
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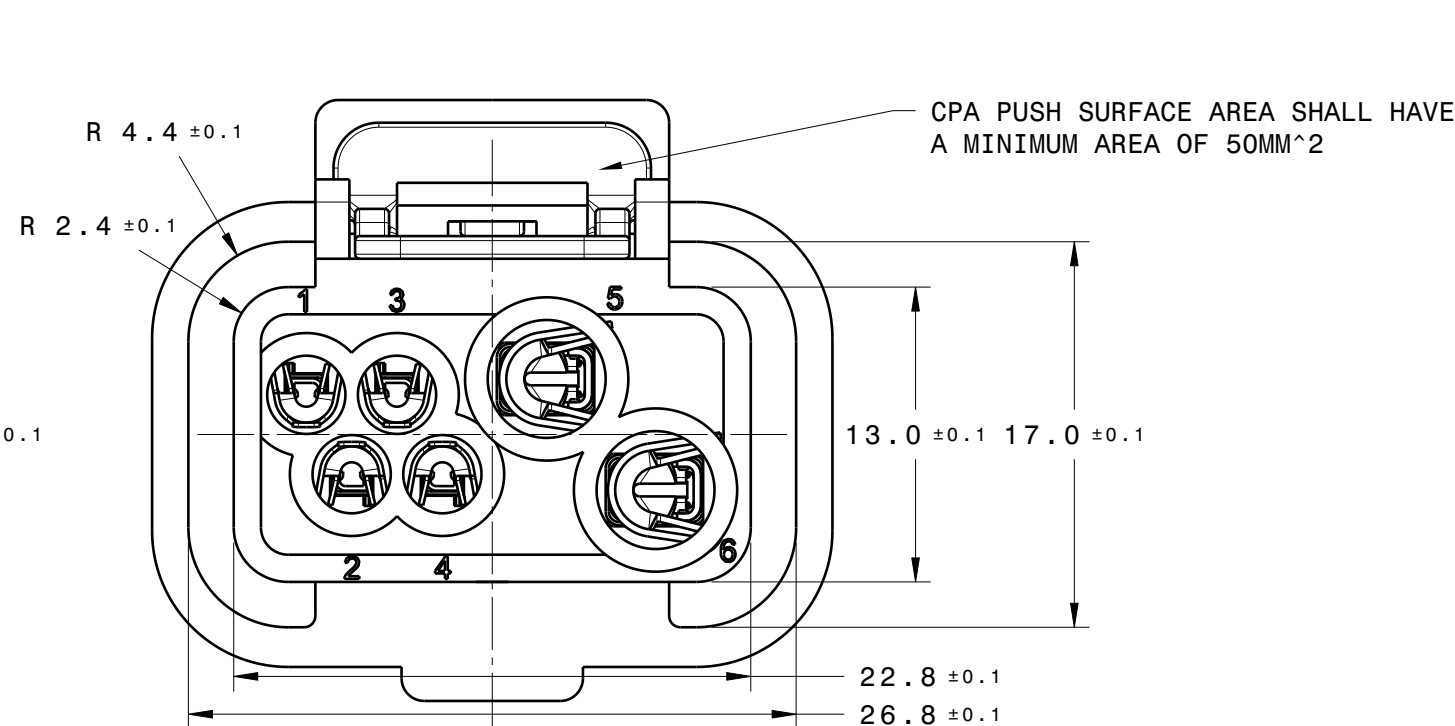
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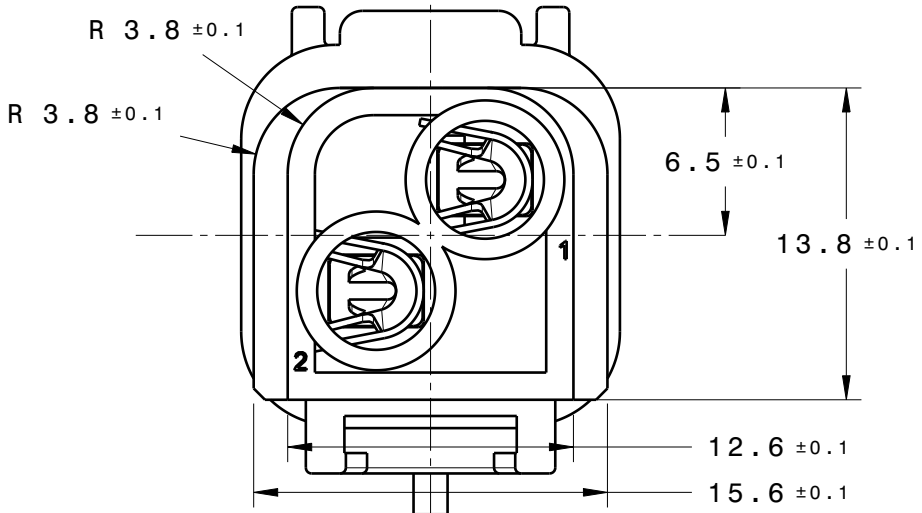
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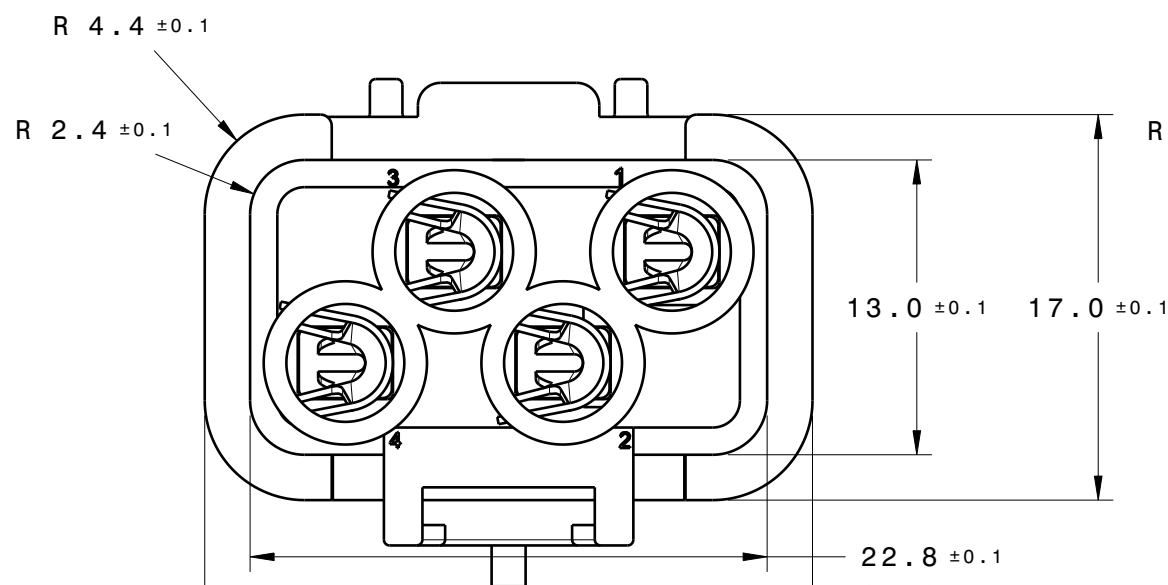
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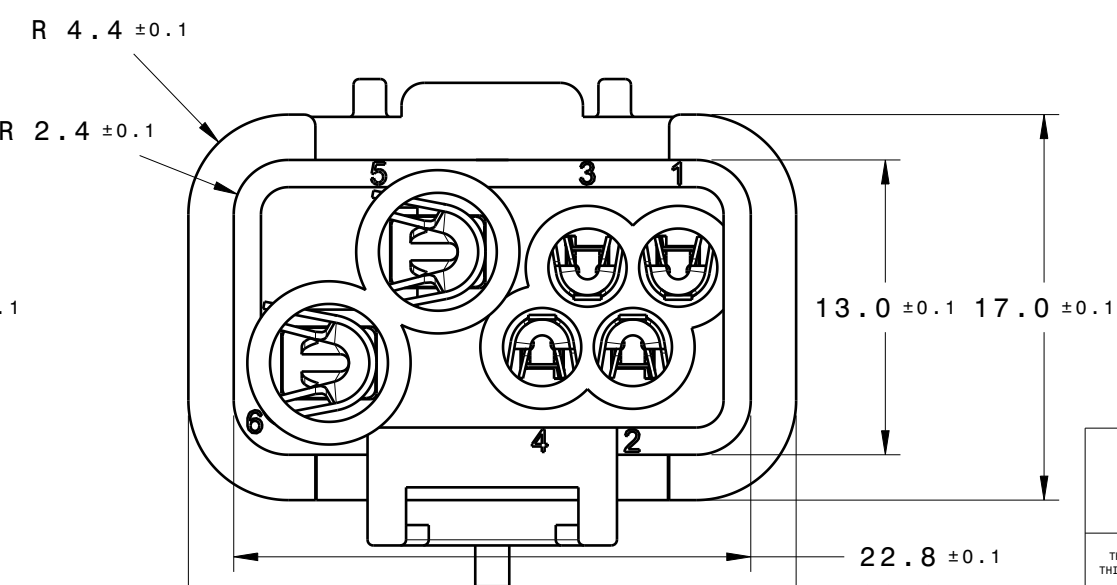
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2280011

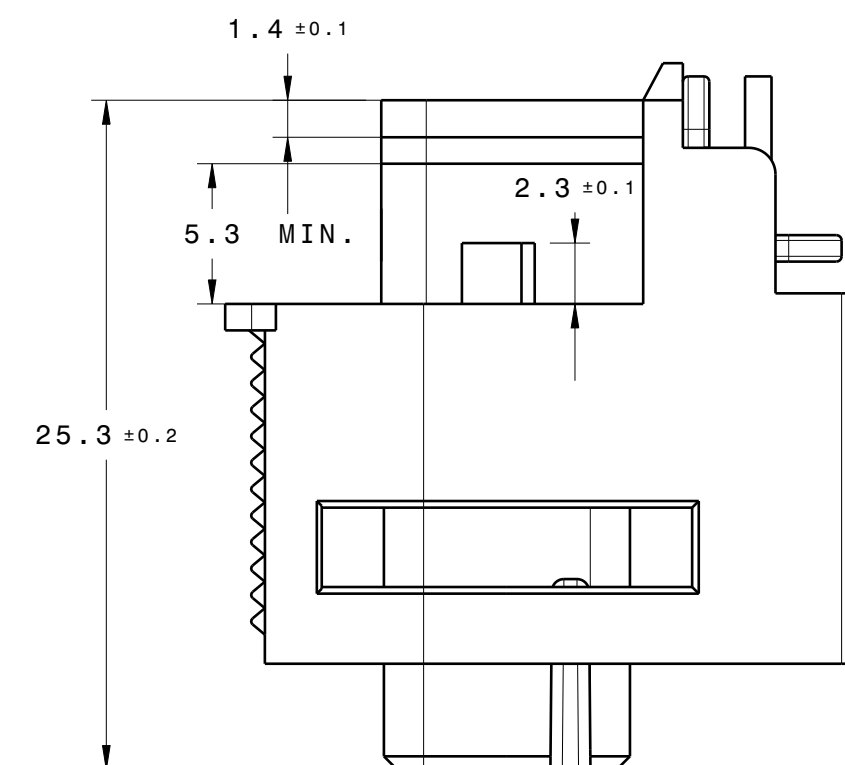
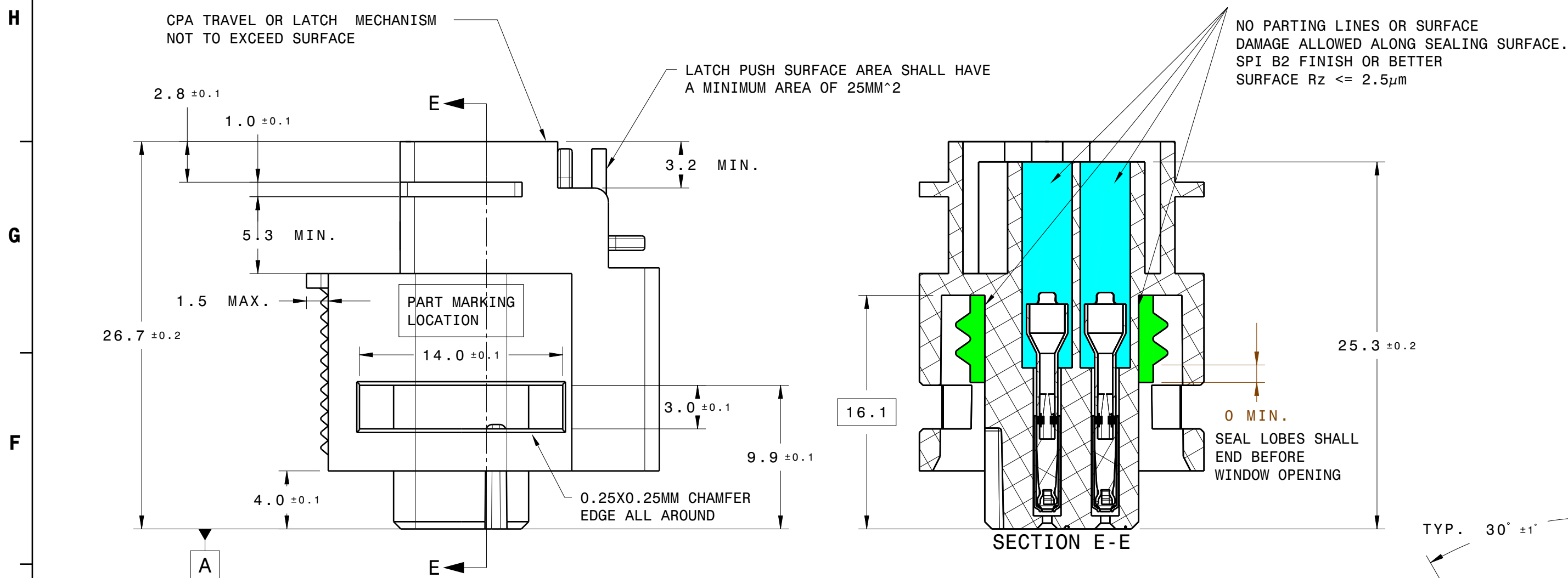


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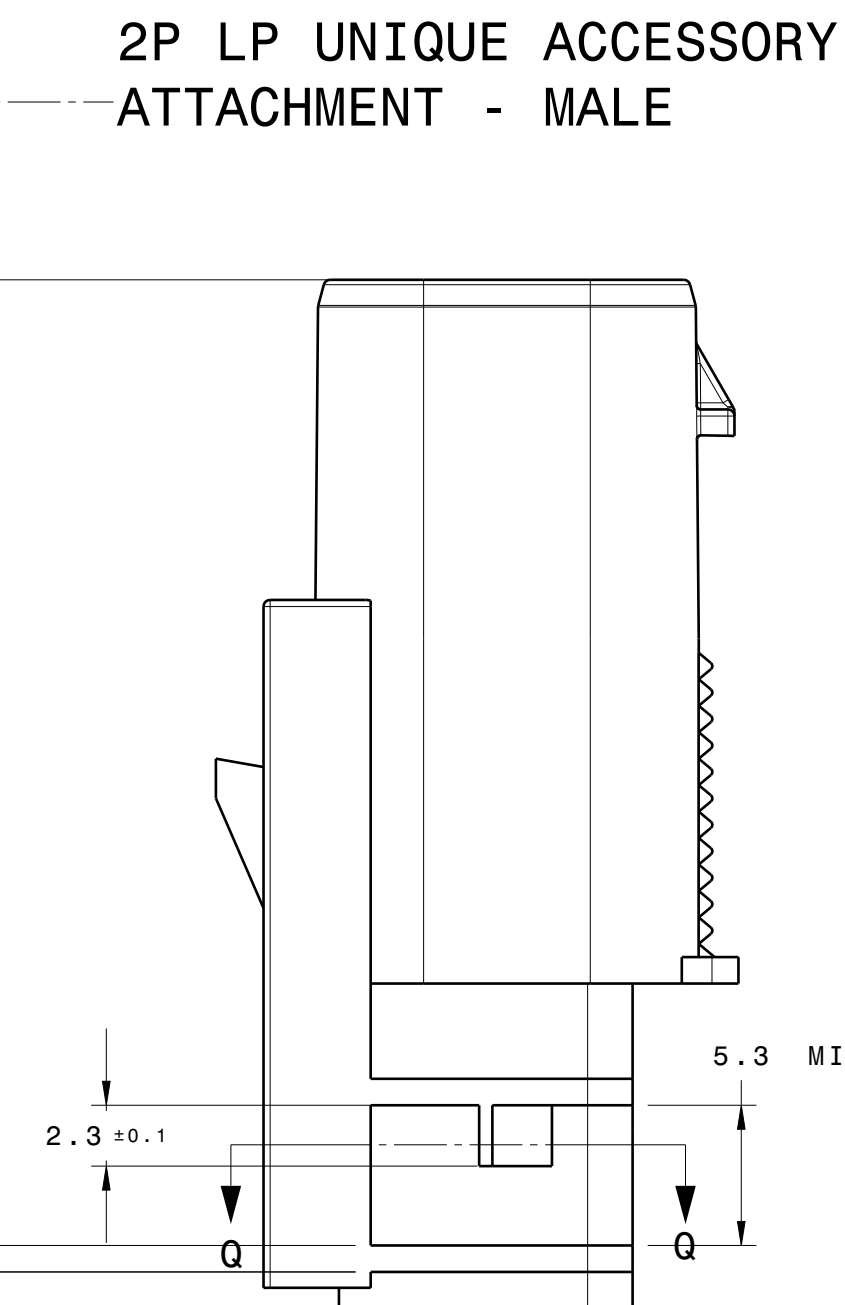
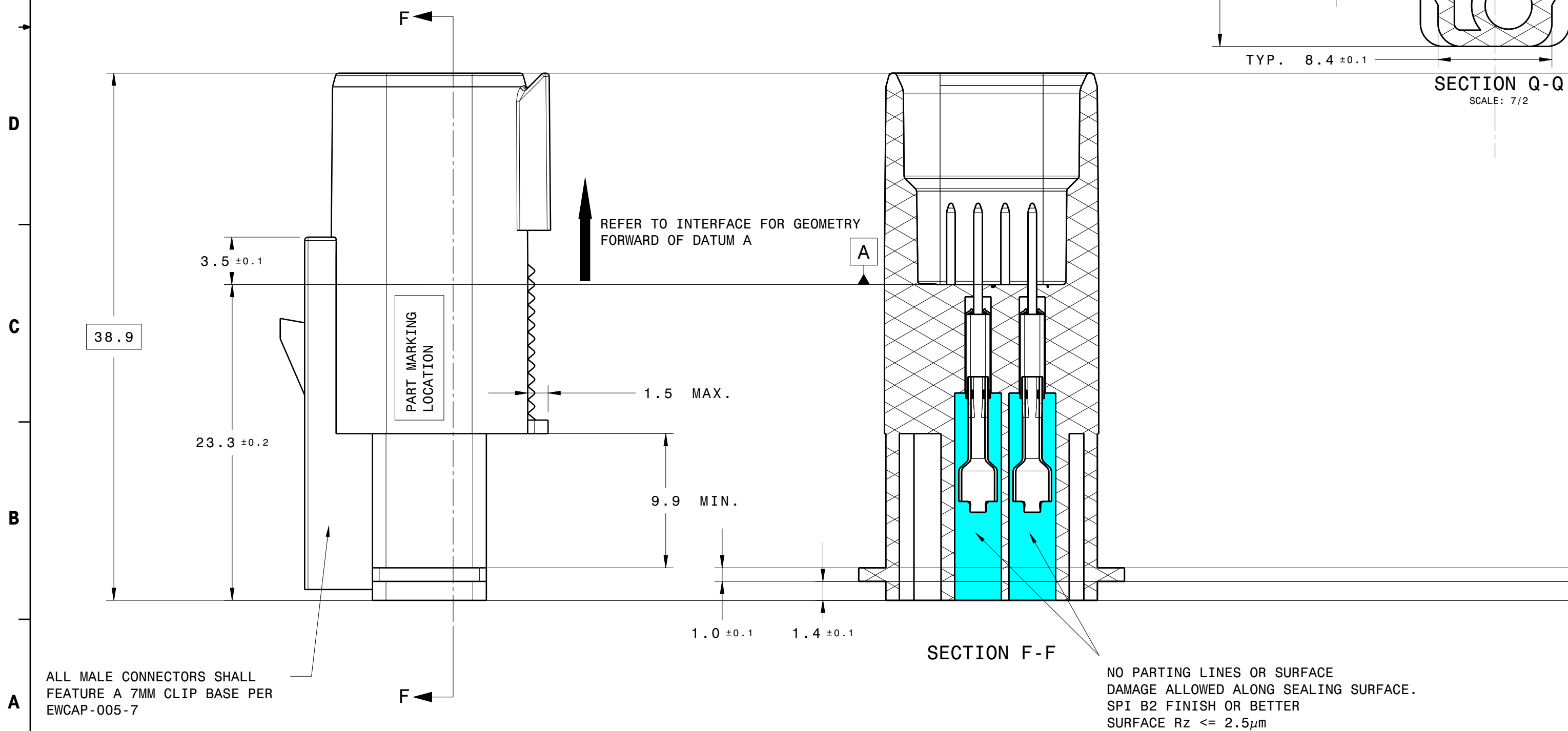


6280011

2P LP UNIQUE ACCESSORY
ATTACHMENT - FEMALE

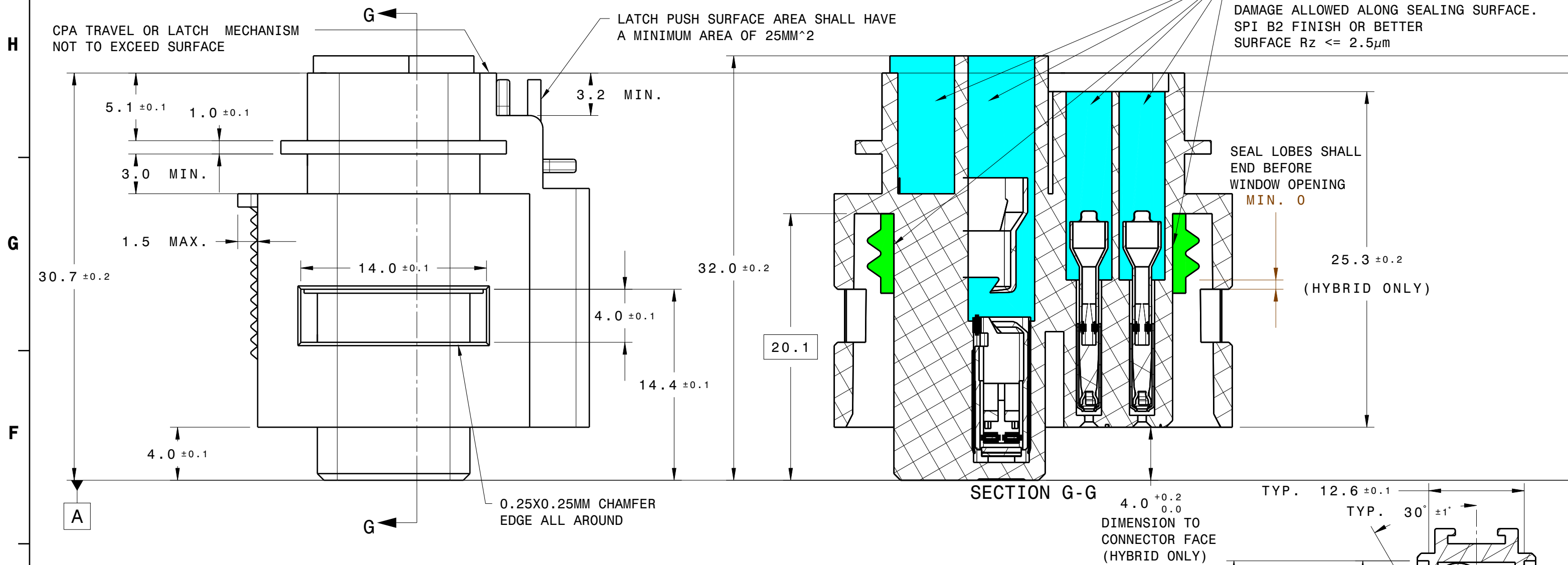


LOW POWER (LP) MALE CONNECTOR TYPICAL DIMENSIONS

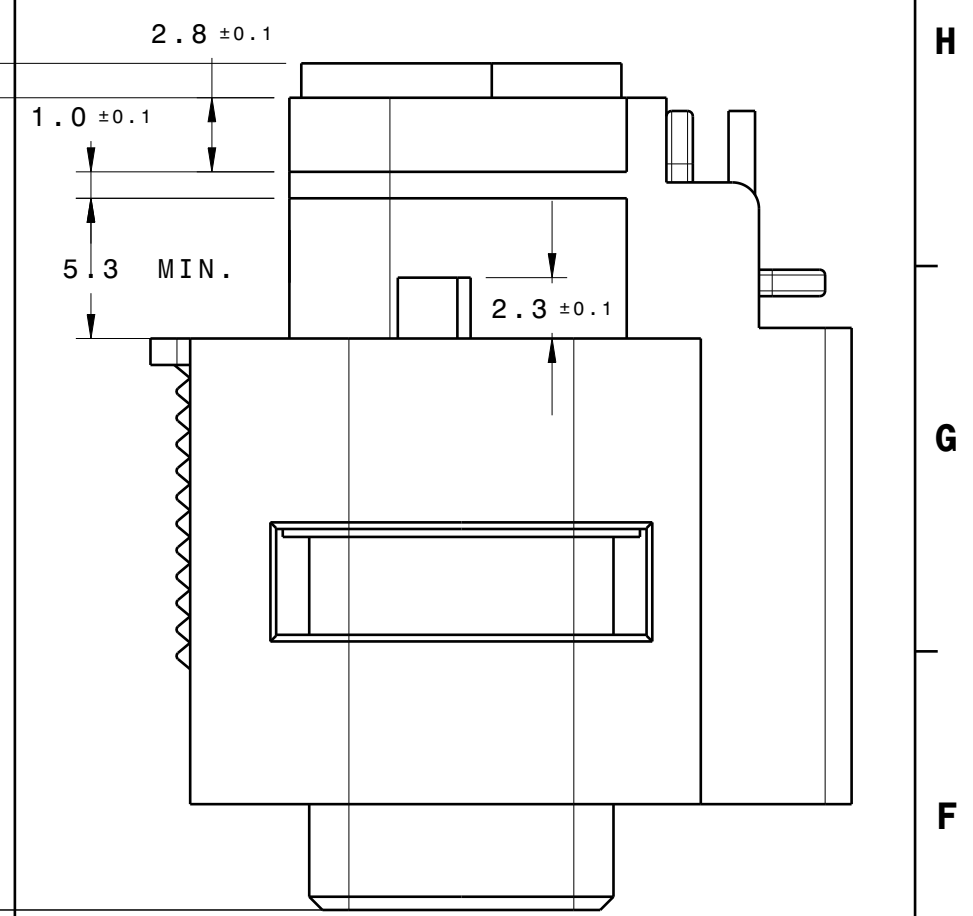


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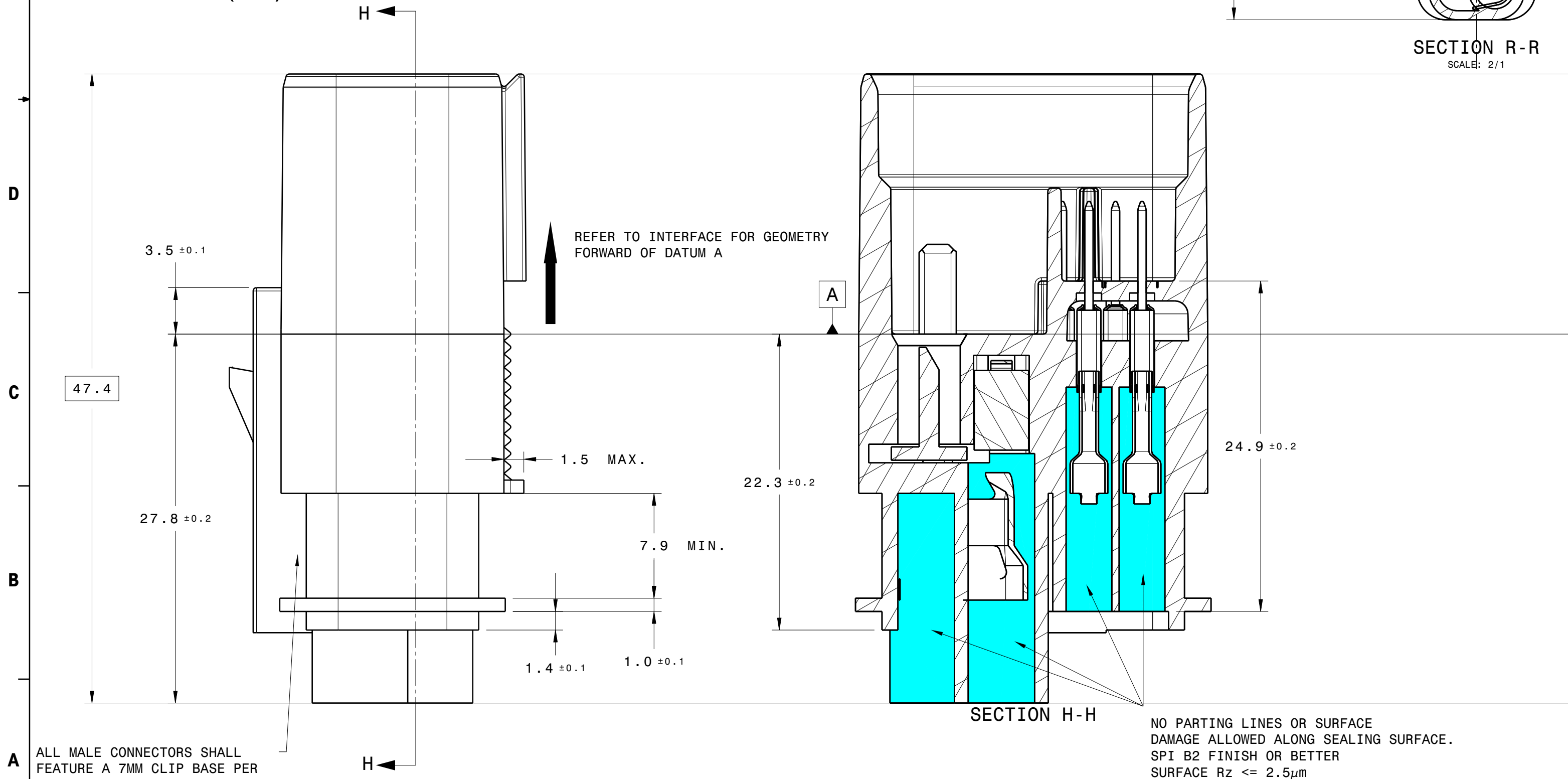
HIGH POWER (HP) AND HYBRID FEMALE CONNECTOR DIMENSIONS



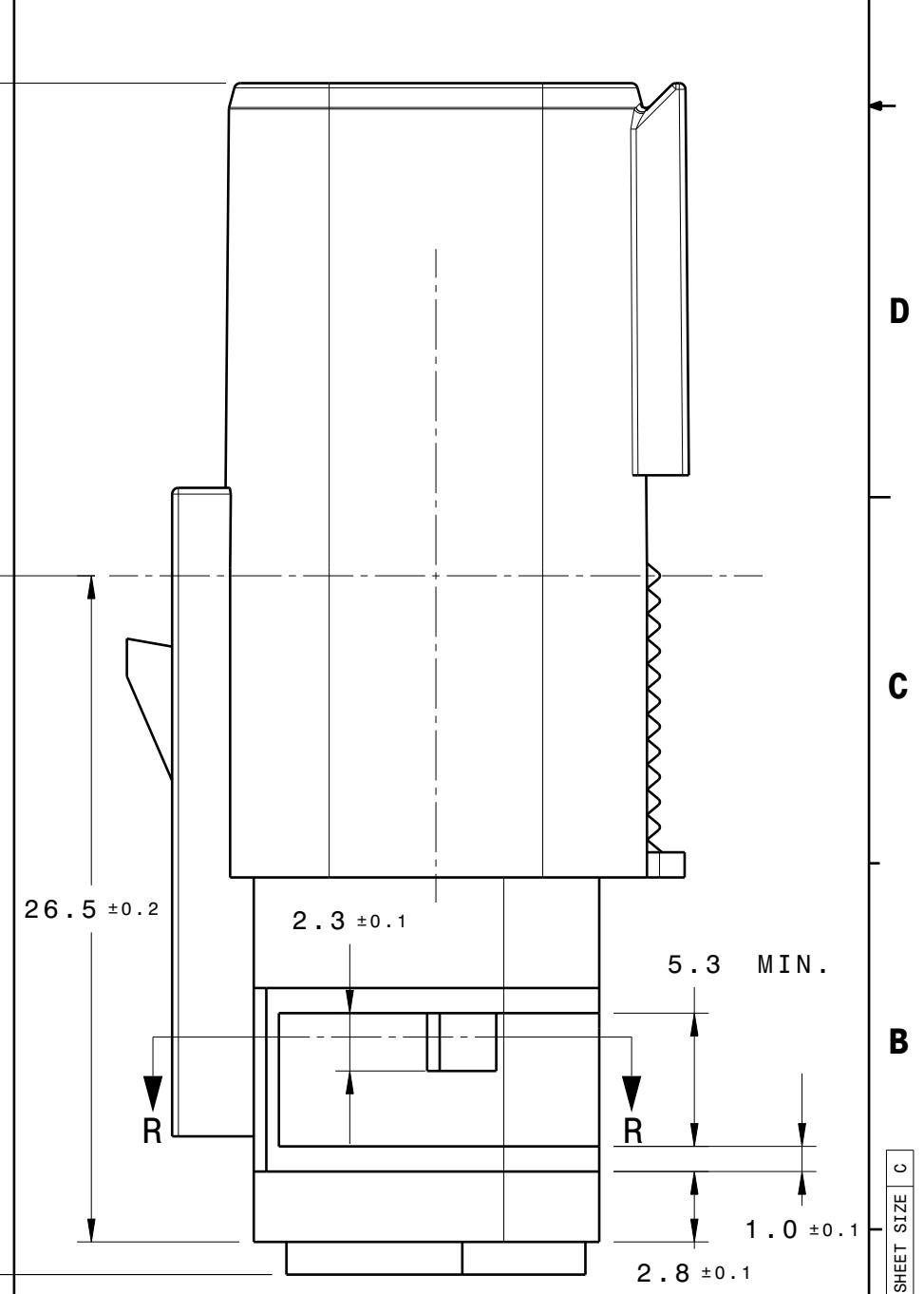
2P HP UNIQUE ACCESSORY ATTACHMENT - FEMALE



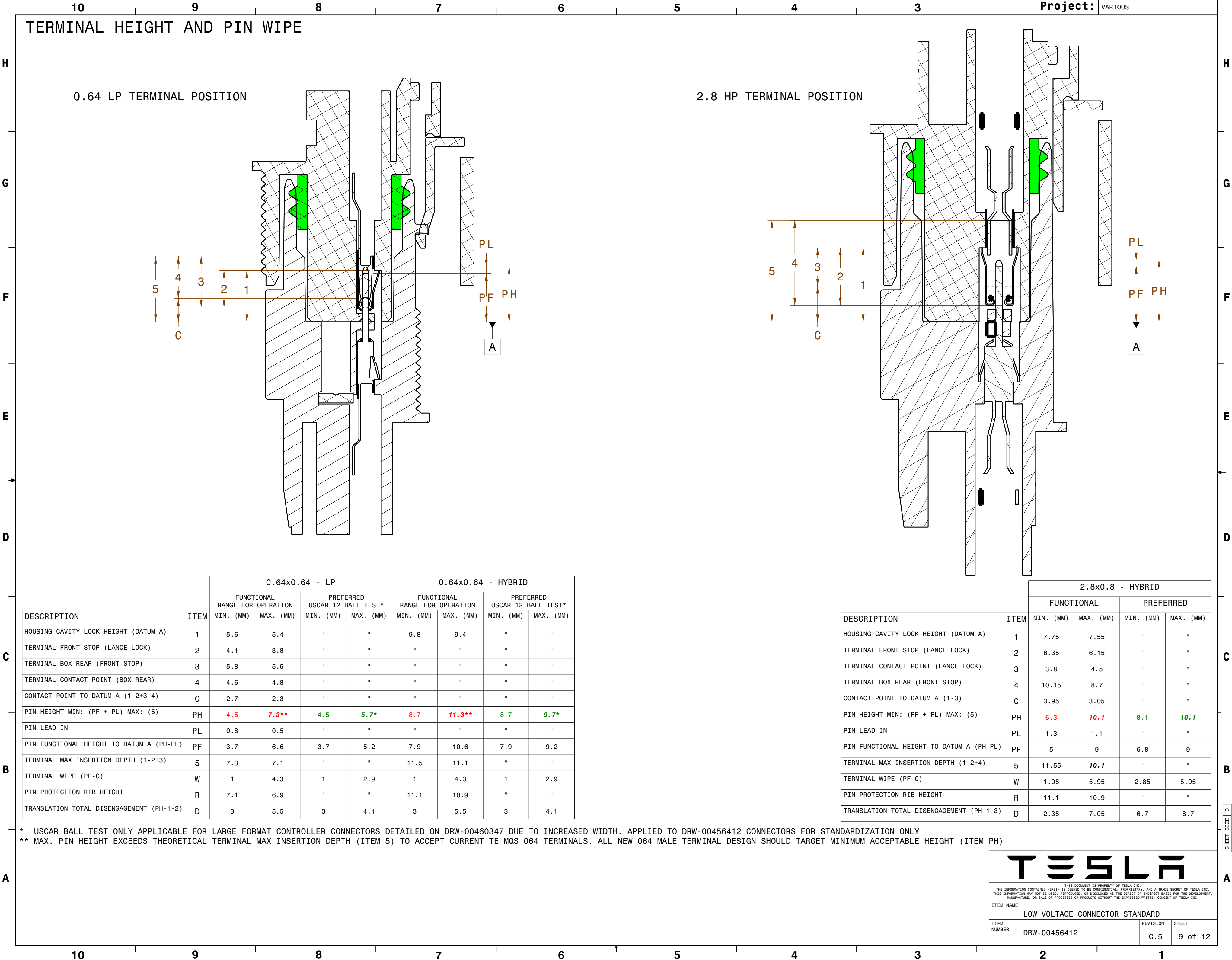
HIGH POWER (HP) AND HYBRID MALE CONNECTOR DIMENSIONS



2P HP UNIQUE ACCESSORY ATTACHMENT - MALE



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ITEM NAME		
LOW VOLTAGE CONNECTOR STANDARD		
ITEM NUMBER	REVISION	SHEET
DRW-00456412	C.5	8 of 12



109876543

Project:VARIOUS

TERMINAL AND SEAL USAGE

064 TERMINALS SHALL BE TE MQS PART NUMBERS SPECIFIED BELOW OR A CAVITY COMPATIBLE EQUIVALENT MEETING ALL SPECIFIED DIMENSIONS IN THIS DOCUMENT AND DEMONSTRATED TO EQUAL OR EXCEED ELECTRICAL AND THERMAL PERFORMANCE. DETAILED INTERNAL CAVITY DIMENSIONS SHALL BE SPECIFIED BY THE TERMINAL SUPPLIER.

280 TERMINALS SHALL BE TE MCP PART NUMBERS SPECIFIED BELOW OR A CAVITY COMPATIBLE EQUIVALENT MEETING ALL SPECIFIED DIMENSIONS IN THIS DOCUMENT AND DEMONSTRATED TO EQUAL OR EXCEED ELECTRICAL AND THERMAL PERFORMANCE. DETAILED INTERNAL CAVITY DIMENSIONS SHALL BE SPECIFIED BY THE TERMINAL SUPPLIER.

COMPATIBLE SEALS AND PLUGS SHALL BE PART NUMBERS SPECIFIED BELOW OR AN EQUIVALENT DEMONSTRATED TO MEET ALL PERFORMANCE REQUIREMENTS SPECIFIED IN THIS DOCUMENT.

064 FEMALE 4 BEAM TERMINALS						
Size	Wire (mm²)	Plating	TE TERMINAL	TE SEAL	TE CAV. PLUG	
064	0.13-0.17	Sn	2141826-1	967067-2	967056-1	
064	0.25-0.35	Sn	5-962885-1			
064	0.5-0.75	Sn	5-965906-1	967067-1		
064	0.13-0.17	Ag	2141826-6	967067-2		
064	0.25-0.35	Ag	5-962885-6			
064	0.5-0.75	Ag	5-965906-6	967067-1		
064	0.13-0.17	Au	2141826-5	967067-2		
064	0.25-0.35	Au	5-962885-5			
064	0.5-0.75	Au	5-965906-5	967067-1		
064 FEMALE 2 BEAM TERMINALS						
Size	Wire (mm²)	Plating	TE TERMINAL	TE SEAL	TE CAV. PLUG	
064	0.13-0.17	Sn	2479123-1	967067-2	967056-1	
064	0.25-0.35	Sn	2479013-1			
064	0.5-0.75	Sn	2479014-1	967067-1		
064	0.13-0.17	Ag	2479123-6	967067-2		
064	0.25-0.35	Ag	2479013-6			
064	0.5-0.75	Ag	2479014-6	967067-1		
064 MALE TERMINALS						
Size	Wire (mm²)	Plating	TE TERMINAL	TE SEAL	TE CAV. PLUG	
064	0.13-0.17	Sn	2141830-1	967067-2	967056-1	
064	0.25-0.35	Sn	5-962886-1			
064	0.5-0.75	Sn	5-965908-1	967067-1		
064	0.13-0.17	Ag	2141830-3	967067-2		
064	0.25-0.35	Ag	5-962886-3			
064	0.5-0.75	Ag	5-965908-3	967067-1		
064	0.13-0.17	Au	2141830-2	967067-2		
064	0.25-0.35	Au	5-962886-2			
064	0.5-0.75	Au	5-965908-2	967067-1		
280 FEMALE TERMINALS - CuNiSi						
Size	Wire (mm²)	Plating	TE TERMINAL	TE SEAL	TE CAV. PLUG	
280	0.2-0.5	Sn	1-2141859-1	828904-1	828922-1	
280	0.5-1	Sn	1-968855-1			
280	1.5-2.5	Sn	1-968857-1	828905-1		
280	12 AWG	Sn	1-1719506-1	282536-1		
280	4	Sn	1-968859-1	TBD		
280	0.2-0.5	Ag	1-2141859-3	828904-1		
280	0.5-1	Ag	1-968855-3			
280	1.5-2.5	Ag	1-968857-3	828905-1		
280	12 AWG	Ag	1-1719506-3	282536-1		
280	4	Ag	1-968859-3	TBD		
280	0.2-0.5	Au	1-2141859-2	828904-1		
280	0.5-1	Au	1-968855-2			
280	1.5-2.5	Au	1-968857-2	828905-1		
280 MALE TERMINALS - CuSn (Default)						
Size	Wire (mm²)	Plating	TE TERMINAL	TE SEAL	TE CAV. PLUG	
280	0.2-0.5	Sn	1-965982-1	828904-1	828922-1	
280	0.5-1	Sn	1-962915-1			
280	1-2.5	Sn	1-962916-1	828905-1		
280	12 AWG	Sn	1-1719504-1	282536-1		
280	2.5-4	Sn	1-968947-1	TBD		
280	0.5-1	Ag	1-962915-2	828904-1		
280	1-2.5	Ag	1-962916-2	828905-1		
280	12 AWG	Ag	1-1719504-2	282536-1		
280	2.5-4	Ag	1-968947-2	TBD		
280	0.2-0.5	Au	1-965982-3	828904-1		
280	0.5-1	Au	1-962915-3			
280	1-2.5	Au	1-962916-3	828905-1		
280 MALE TERMINALS - CuFe (Option)						
Size	Wire (mm²)	Plating	TE TERMINAL	TE SEAL	TE CAV. PLUG	
280	0.5-1	Sn	2-962915-1	828904-1	828922-1	
280	1-2.5	Sn	2-962916-1	828905-1		
280	2.5-4	Sn	2-968947-1	TBD		
280	0.5-1	Ag	2-962915-2	828904-1		
280	1-2.5	Ag	2-962916-2	828905-1		
280	2.5-4	Ag	2-968947-2	TBD		
280	0.5-1	Au	2-962915-3	828904-1		
280	1-2.5	Au	2-962916-3	828905-1		

PERFORMANCE REQUIREMENTS

AT A MINIMUM, ALL CONNECTORS ARE EXPECTED TO MEET APPLICABLE PERFORMANCE REQUIREMENTS OF USCAR-2

THE FOLLOWING CLASSIFICATIONS ARE TO BE USED:

TEMPERATURE CLASS: T2
SEALING CLASS: S2
S3 WHEN BACKSHELL ACCESSORY FITTED

VIBRATION CLASS: V3 IN SEALED CONFIGURATION
V1 IN UNSEALED CONFIGURATION

ERGONOMIC CLASS: USCAR-25 CLASSIFICATION NOT REQUIRED
SEALED CONNECTOR MATING FORCE SHALL BE EQUAL TO OR BELOW 75N

PARTS SUPPLIED TO TESLA MUST MEET ALL ADDITIONAL PERFORMANCE REQUIREMENTS LISTED IN THE GENERAL NOTES OF THIS DOCUMENT.

GENERAL REQUIREMENTS

SEALING:
WIRE SEALS AND HOUSING RADIAL SEAL SHALL BE AN OPTIONAL PART. CONNECTOR SHALL BE FUNCTIONAL WITHOUT SEALS INSTALLED

CODING:
FOUR KEY CODES (A,B,C,D) PLUS UNIVERSAL CODE (0) OPTIONS SHALL BE AVAILABLE

CONNECTOR POSITION ASSURANCE (CPA):
CPA SHALL BE AVAILABLE AS AN OPTIONAL PART

CYCLE RATING:
10 CYCLES MINIMUM FOR TIN PLATED TERMINALS (DRY/UNLUBRICATED)
50 CYCLES MINIMUM FOR SILVER PLATED TERMINALS (DRY/UNLUBRICATED)

HOUSING COLOR:
0V-16V - BLACK
16V 60V - LIGHT BLUE (RAL 5012)

CREEPAGE AND CLEARANCE:
ALL CONNECTORS SHALL PASS IEC-60664-1 CREEPAGE AND CLEARANCE REQUIREMENTS FOR THE APPLICABLE VOLTAGES WHILE MEETING:
BASIC INSULATION RATING
ALTITUDE = 5000M (CORRECTION FACTOR = 1.48)
POLLUTION DEGREE: 3

	MATERIAL GROUP I (RECOMMENDED)	MATERIAL GROUP II	MATERIAL GROUP III
MIN. CREEPAGE (MM)	1.6	1.8	2.0
MIN. CLEARANCE (MM)	1.2	1.2	1.2

MATERIALS:
ALL PLASTICS SHALL MEET HB FLAMMABILITY PER UL94 OR GREATER
ELECTRICAL RTI MUST MEET 125C OPERATING TEMPERATURE
MATERIAL RATED TO CTI (COMPARATIVE TRACKING INDEX) EQUAL TO 600 OR GREATER, PLC = 0, UL840 MATERIAL GROUP 1 IS RECOMMENDED

SIGNAL INTEGRITY:
CONNECTORS SHALL MEET 100BASE-TX AND 10BASE-T1S WITH TWISTED PAIR WIRING

TEST REQUIREMENTS

ISOLATION RESISTANCE:
CONNECTORS SHALL MEET A MINIMUM ISOLATION RESISTANCE OF 100MOHM AT 330VDC BETWEEN ADJACENT CONDUCTORS

HIPOT TESTING:
CONNECTORS SHALL WITHSTAND A TEST VOLTAGE BETWEEN ADJACENT CONDUCTORS RAISED UNIFORMLY FROM 0V TO 330VDC AT 23C +/-5C THERE SHALL BE NO DIELECTRIC BREAKDOWN OR FLASH OVER BETWEEN CAVITIES OR TO THE OUTSIDE AT ANY TIME DURING THE TEST.

PRESSURE TEST (SEALED ONLY):
DURING ANY CONNECTOR LEVEL PRESSURE TEST, FIXTURE MUST SEAL OFF ALL SEALING SURFACES (HEADER SIDE/WIRE SEAL BORES OR FEMALE SIDE SEAL). TPA/ISL MUST NOT BE CYCLED.

DURING ANY HARNESS LEVEL PRESSURE TEST, FIXTURE ONLY SHOULD FORM THE HEADER SIDE SEALING SURFACE (PER TS INTERFACE).

NO DAMAGE TO ANY SEALING SURFACES FROM FIXTURE PERMITTED.

TEST PRESSURE: +1.5PSI
ACCEPTANCE CRITERIA: <0.03PSI PRESSURE LOSS
TEST TIME: 3 SECONDS
TEST FIXTURE VOLUME INCLDING TUBING: <1.8E-5 M^3

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ITEM NAME
LOW VOLTAGE CONNECTOR STANDARD

ITEM NUMBER
DRW-00456412

REVISION
C.5

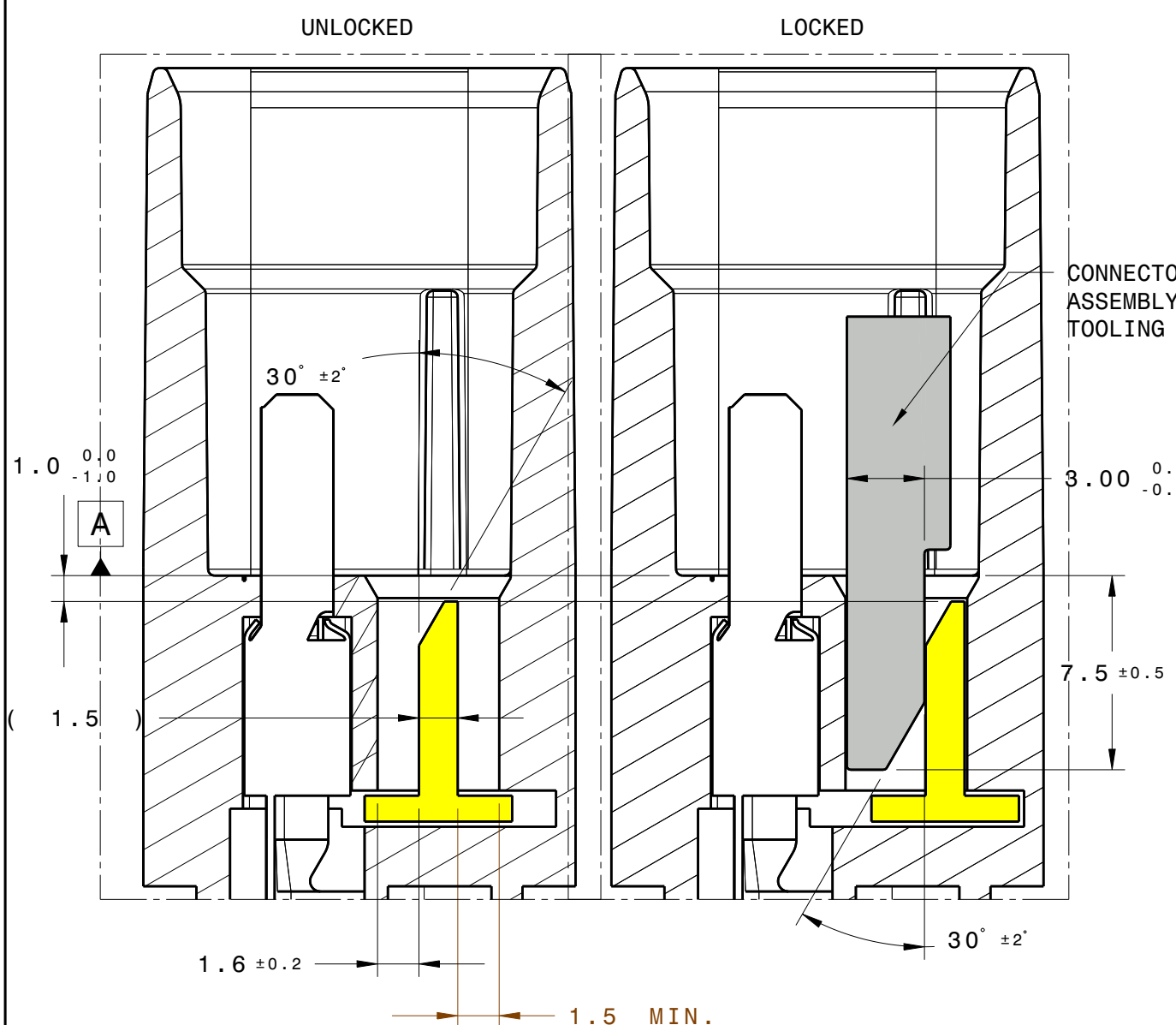
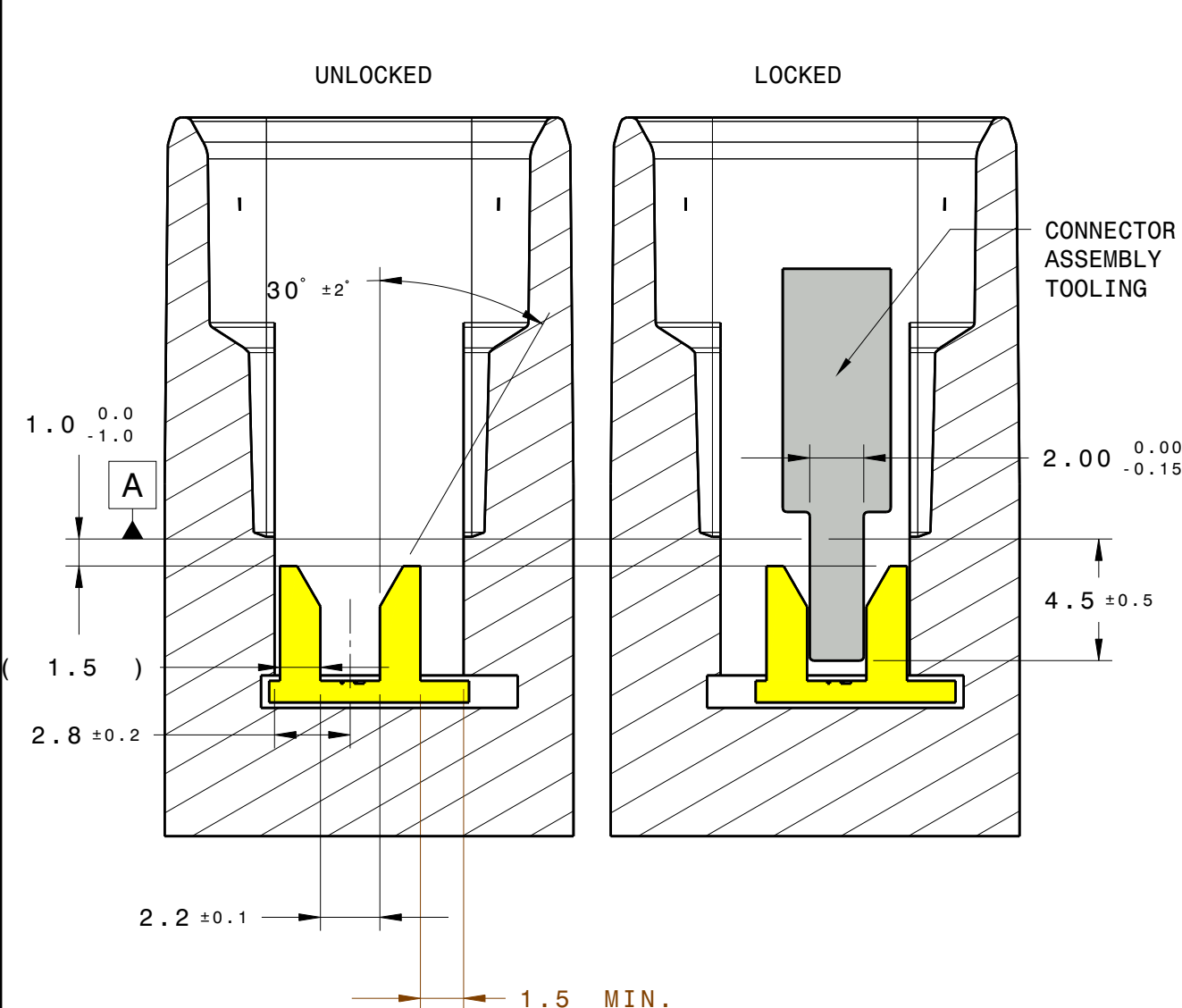
SHEET
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109876543

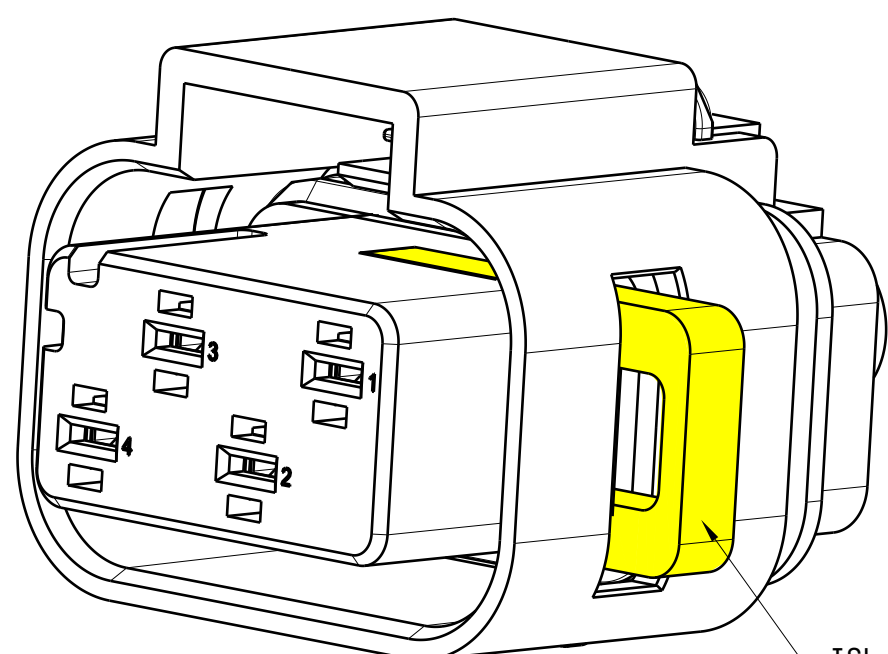
21

ADDITIONAL REQUIREMENTS (ALL):
TRAVEL TO LOCK: 1.5MM MIN.
CONNECTOR TPA LOCKING FORCES: PER USCAR-2
CONNECTOR/TPA OVER TRAVEL FORCE WITHSTAND: 65N
CONNECTOR ASSEMBLY TOOLING APPLIED FORCE: 60N MIN. - 65N MAX

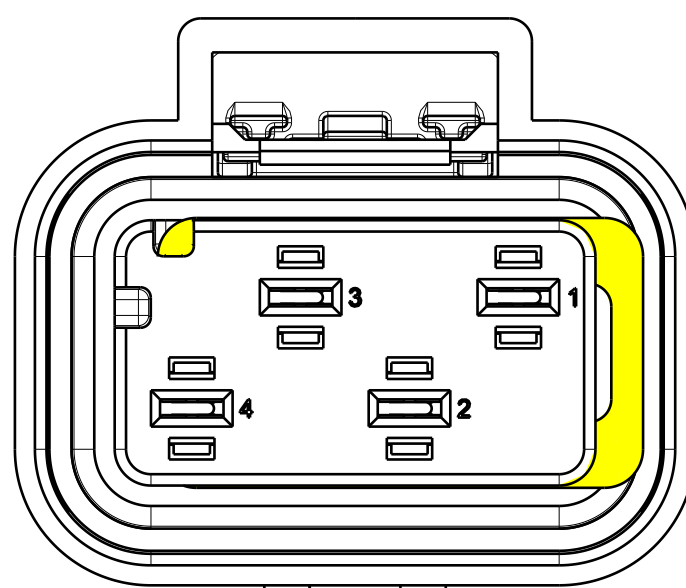
TPA TYPE 3 DETAIL



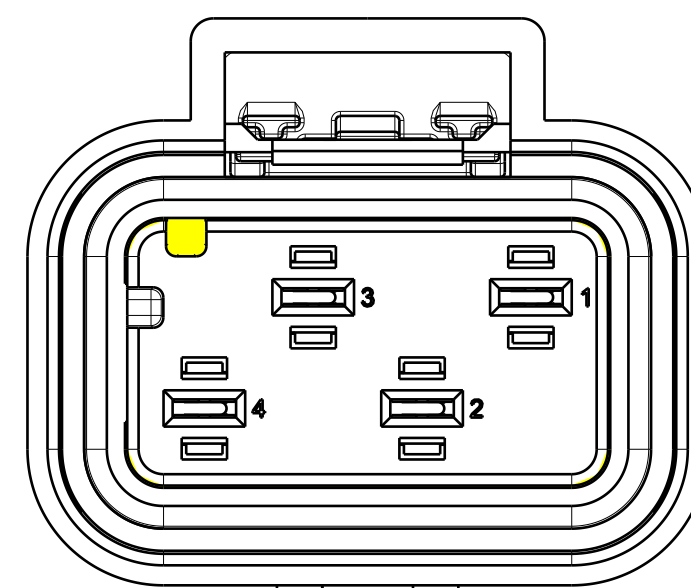
FEMALE ISL (INDEPENDANT SECONDARY LOCK)



ADDITIONAL REQUIREMENTS (ALL):
CONNECTOR TPA LOCKING FORCES: PER USCAR-2
CONNECTOR/TPA OVER TRAVEL FORCE WITHSTAND: 65N
CONNECTOR ASSEMBLY TOOLING APPLIED FORCE: 60N MIN. - 65N MAX



UNLOCKED POSITION
(EXAMPLE ONLY)
SCALE: 3/1



LOCKED POSITION
(EXAMPLE ONLY)
SCALE: 3/1

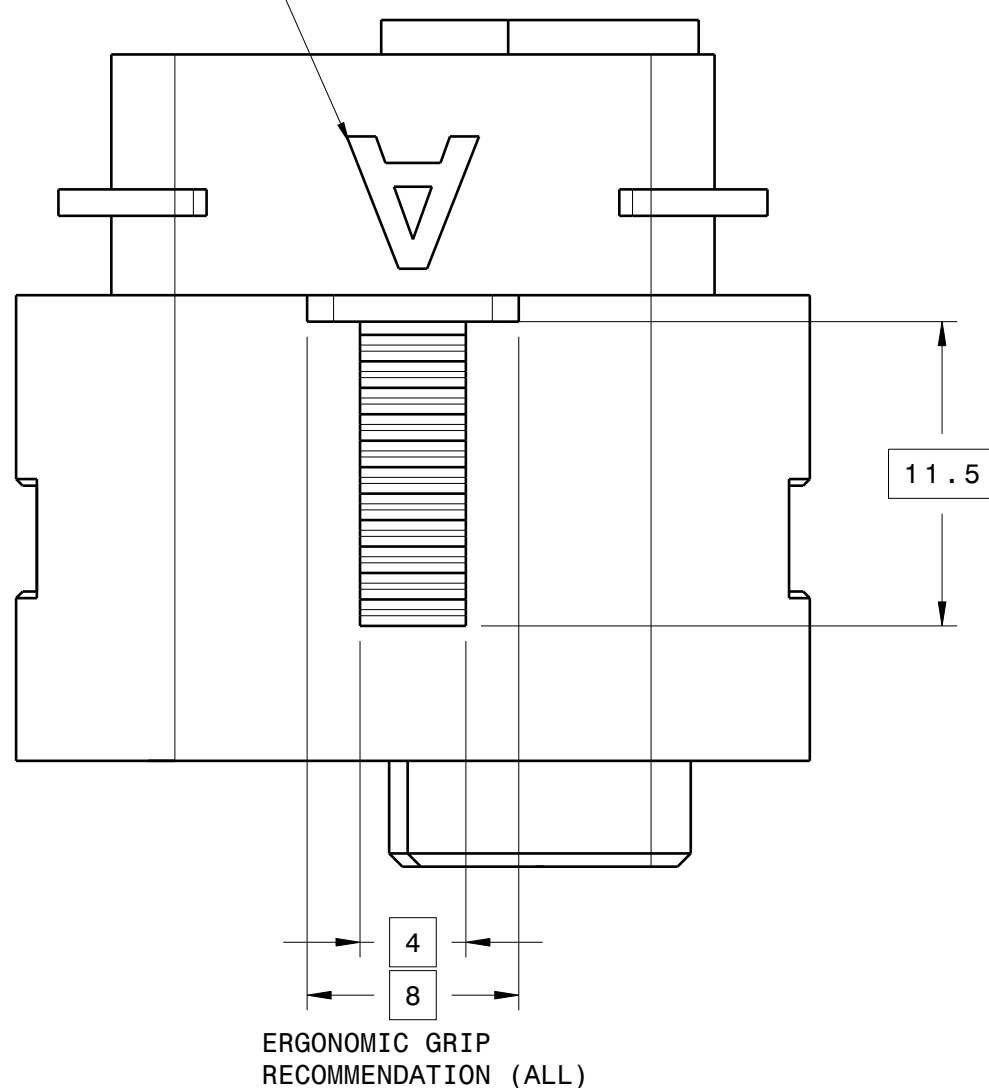
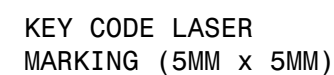


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MANUFACTURE, OR SALE OF PROCESSES OR PRODUCTS WITHOUT THE EXPRESSED WRITTEN CONSENT OF TESLA INC.

ITEM NAME	LOW VOLTAGE CONNECTOR STANDARD
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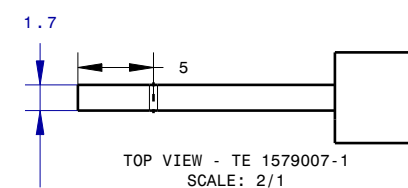
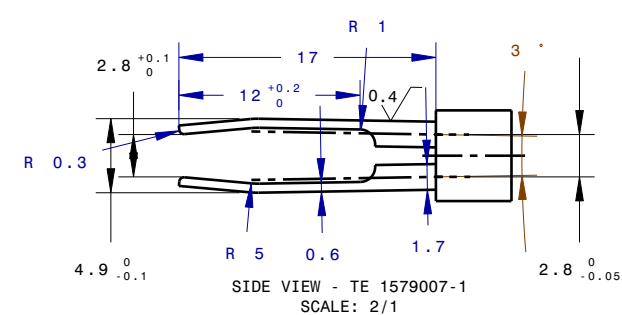
ITEM NUMBER	DRW-00456412	REVISION	SHEET
		C.5	11 of 1

KEY MARKING

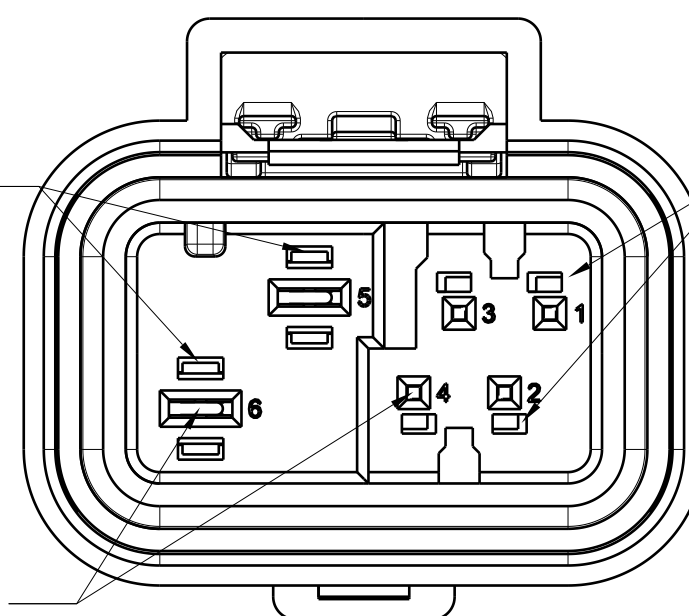


ELECTRICAL PROBING AND TERMINAL REMOVAL

POGO PIN/EXTRACTION TOOL/PROBING TOOL MUST NOT INSERT INTO TERMINAL SLIDING CONTACT



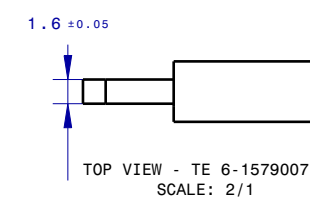
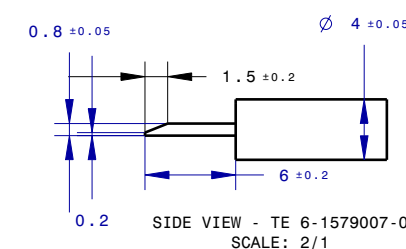
OPENING FOR TERMINAL EXTRACTION (LOCKING LANCE)
AND MANUAL ELECTRICAL PROBING.
OPENING MUST BE COMPATIBLE WITH TE 1579007-1 OR
EQUIVALENT.



NO TOOLS ALLOWED
TO AVOID DAMAGE
TO SLIDING
CONTACT

6280012-**-B
SHOWN FOR REFERENCE
SCALE: 1/1

- OPENING FOR TERMINAL EXTRACTION (LOCKING LANCE)
AND ELECTRICAL PROBING.
OPENING MUST BE COMPATIBLE WITH TE 6-1579007-0 OR
EQUIVALENT.



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ITEM NAME LOW VOLTAGE CONNECTOR STANDARD		
ITEM NUMBER DRW-00456412	REVISION C.5	SHEET 12 of 12